



Intelligent Systems

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Lab 1

Bayes Nets in pomegranate

Learning objectives for this week are:

1. to be able to install the `pomegranate` Python package
2. to get familiar with `jupyter` notebooks

1.1 Installing pomegranate

Follow the steps below¹. If one of the commands in the tutorials does not work, you usually need to install some Python packages. We'll use Python 3, so you'll have to make sure you use the appropriate version of `pip`. (if in doubt, just use `python -m pip` instead of `pip`).

1. Install Anaconda for your username, by following the instructions in this tutorial.

Please browse it even if you already have `conda` installed on your machine.

2. Update `conda` to the last version:

```
conda update -n base -c defaults conda
```

3. Create a `conda` environment, say `johndoe`

```
conda create --name johndoe python=3.7.2
```

(see also the tutorial mentioned at step #1 on how to do that)

4. Activate (i.e., start) your own environment (here it is called `johndoe`)

```
conda activate johndoe
```

(to deactivate it, you can do: `conda deactivate`)

5. install Python 3.7.2:

```
conda install -c anaconda python=3.7.2
```

6. Install the Jupyter Notebook App:

```
pip install jupyter
```

¹Tested on Ubuntu 18.04 LTS, Python 3.7.2

7. Install the `pomegranate` package:

```
pip install pomegranate==0.11.0
```

8. get the appropriate `jupyter` notebook from the following tutorial

- (a) right click on `B.Model.Tutorial_4.Bayesian_Networks.ipynb`
- (b) click the `Raw` button on top right part
- (c) save it (`Ctrl-S`)

9. (this step is usually needed if you have issues with `pygraphviz` –see below). Check that you have `graphviz` installed on you system, by typing `dot -V` at your bash prompt. If not, you'll have to do the following steps (you need to be a sudoer):

```
sudo apt install graphviz
```

```
sudo apt install graphviz-dev
```

```
sudo apt install --reinstall build-essential
```

10. From inside the `conda` environment, start `jupyter` by typing

```
jupyter notebook
```

If one of the imports fails, you'll have to do a `pip install`. For example, in order to be able to run the tutorial mentioned in Section ??, we had to perform:

```
pip install matplotlib
```

```
pip install seaborn
```

```
pip install watermark
```

```
pip install pygraphviz
```

```
pip install sklearn
```

```
pip install Pillow
```

```
pip install fancyimpute
```

11. Install/upgrade `numpy`:

```
pip install --upgrade numpy==1.20.1
```